

# 1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

## I. Data type of all columns in the "customers" table

Output Screenshot:

| Field name               | Type    |
|--------------------------|---------|
| customer_id              | STRING  |
| customer_unique_id       | STRING  |
| customer_zip_code_prefix | INTEGER |
| customer_city            | STRING  |
| customer_state           | STRING  |

Insights:

- Customers table consist of customer identification details like customer\_id & customer\_unique\_id
- Also includes customer's geographical data like zip\_code\_prefix, city and state
- All these details are stored by utilizing both STRING and INTEGER data types

## II. Get the time range between which the orders were placed.

Query:

```
select min(order_purchase_timestamp) as Earliest_order_timestamp,
       max(order_purchase_timestamp) as Latest_order_timestamp
from `target.orders`
```

Output screenshot:

| Row | Earliest_order_timestamp | Latest_order_timestamp  |
|-----|--------------------------|-------------------------|
| 1   | 2016-09-04 21:15:19 UTC  | 2018-10-17 17:30:18 UTC |

Insights:

- All orders were placed between 2016-09-04 21:15:19 UTC to 2018-10-17 17:30:18 UTC

## III. Count the Cities & States of customers who ordered during the given period.

Query:

```
select count(distinct a.customer_city) as order_city_count,
       count(distinct a.customer_state) as order_state_count
from `target.customers` a inner join `target.orders` b on
a.customer_id=b.customer_id
```

### Output screenshot:

| Row | order_city_count | order_state_count |
|-----|------------------|-------------------|
| 1   | 4119             | 27                |

### Insights:

- Orders were placed from 4119 different cities which are distributed over 27 different states.
- This suggests a broad distribution of orders within a relatively limited number of states.
- It also highlights an opportunity to expand into more states.

### Action plan:

- Check for expansion opportunities in more states and cities
- Can offer state-wise promotions in case of festival, holidays etc.

## 2. In-depth Exploration:

### I. Is there a growing trend in the no. of orders placed over the past years?

#### Query:

```
select extract(year from order_purchase_timestamp) as Order_year,
       count(*) as Order_count
from `target.orders`
group by 1
order by 1
```

### Output screenshot:

| Row | order_year | Order_count |
|-----|------------|-------------|
| 1   | 2016       | 329         |
| 2   | 2017       | 45101       |
| 3   | 2018       | 54011       |

### Insights:

- Prominent trend with substantial increase in order counts year on year.
- Explosive growth in 2017, with the leap from 329 orders in 2016 to 45,101 orders in 2017.
- In year 2018, around 20% growth in orders as compared to previous year 2017

### Action plan:

- Analyze which factors contributed most to the 2017 growth spike and replicate, improve strategies for customer retention.

## II. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

### Query:

```
select format_date('%B - %Y',order_purchase_timestamp) as Order_month,
       count(*) as Order_count
from `target.orders`
group by 1,
       extract(month from order_purchase_timestamp),
       extract(year from order_purchase_timestamp)
order by extract(month from order_purchase_timestamp),
       extract(year from order_purchase_timestamp)
```

### Output screenshot:

| Row | Order_month     | Order_count |
|-----|-----------------|-------------|
| 1   | January - 2017  | 800         |
| 2   | January - 2018  | 7269        |
| 3   | February - 2017 | 1780        |
| 4   | February - 2018 | 6728        |
| 5   | March - 2017    | 2682        |
| 6   | March - 2018    | 7211        |
| 7   | April - 2017    | 2404        |
| 8   | April - 2018    | 6939        |
| 9   | May - 2017      | 3700        |
| 10  | May - 2018      | 6873        |

### Insights:

- Clear spike in orders towards the end of the year 2017, particularly in November and December.
- This upper trend was also carried into early months of 2018.
- Midyear order rate is moderate.

### Action plan:

- Since midyear demand is moderate, run midyear sales events to fill the gap.
- Boost inventory & logistics capacity before November, December to meet the demand spike.

**III. During what time of the day, do the Brazilian customers mostly place their orders?  
(Dawn, Morning, Afternoon or Night)**

- 0-6 hrs: Dawn
- 7-12 hrs: Mornings
- 13-18 hrs: Afternoon
- 19-23 hrs: Night

Query:

```
select case when extract(hour from order_purchase_timestamp) between 0
and 6 then 'Dawn'
        when extract(hour from order_purchase_timestamp) between 7 and 12
then 'Mornings'
        when extract(hour from order_purchase_timestamp) between 13 and
18 then 'Afternoon'
        when extract(hour from order_purchase_timestamp) between 19 and
23 then 'Night' end as Time_of_the_day,
        count(*) as Order_count
from `target.orders`
Group by 1
order by case when Time_of_the_day = 'Dawn' then 1
        when Time_of_the_day = 'Mornings' then 2
        when Time_of_the_day = 'Afternoon' then 3
        when Time_of_the_day = 'Night' then 4 end
```

Output Screenshot:

| Row | Time_of_the_day | Order_count |
|-----|-----------------|-------------|
| 1   | Dawn            | 5242        |
| 2   | Mornings        | 27733       |
| 3   | Afternoon       | 38135       |
| 4   | Night           | 28331       |

Insights:

- Afternoon is the peak period for orders, with 38,135 orders.
- Night and mornings have high order volumes with similar range but lower than afternoon.
- Dawn has the lowest order count, with only 5,242 orders.
- The data shows a significant increase in order volume from Dawn to Mornings, peaks in the afternoon and then a slight decrease during the night.

Action plan:

- Introduce exclusive discounts to encourage activity in the dawn window.
- Ensure higher server uptime for large number of orders in peak hour like afternoon.

### 3. Evolution of E-commerce orders in the Brazil region:

#### I. Get the month-on-month no. of orders placed in each state.

Query:

```
select a.customer_state,
       format_date('%B - %Y',b.order_purchase_timestamp) as Order_month,
       count(b.order_id) as Order_count
from `target.customers` a inner join `target.orders` b on
a.customer_id=b.customer_id
group by 1,2,
        extract(month from b.order_purchase_timestamp),
        extract(year from b.order_purchase_timestamp)
order by 1,
        extract(year from b.order_purchase_timestamp),
        extract(month from b.order_purchase_timestamp)
```

Output Screenshot:

| Row | customer_state | Order_month      | Order_count |
|-----|----------------|------------------|-------------|
| 1   | AC             | January - 2017   | 2           |
| 2   | AC             | February - 2017  | 3           |
| 3   | AC             | March - 2017     | 2           |
| 4   | AC             | April - 2017     | 5           |
| 5   | AC             | May - 2017       | 8           |
| 6   | AC             | June - 2017      | 4           |
| 7   | AC             | July - 2017      | 5           |
| 8   | AC             | August - 2017    | 4           |
| 9   | AC             | September - 2017 | 5           |
| 10  | AC             | October - 2017   | 6           |

Insights:

- SP, RJ, MG, RS and PR are the top 5 states by order count contributing more than 70% of total orders.
- SP has more order volume, nearly 3x more than the next state RJ.
- RR,AC,AP,TO are low volume states.

Action plan:

- Drive multiple marketing strategies in low volume states.

#### II. How are the customers distributed across all the states?

Query:

```
select customer_state,
       count(distinct customer_unique_id) as Total_customers
from `target.customers`
group by 1
```

### Output Screenshot:

| Row | customer_state | Total_customers |
|-----|----------------|-----------------|
| 1   | AC             | 77              |
| 2   | AL             | 401             |
| 3   | AM             | 143             |
| 4   | AP             | 67              |
| 5   | BA             | 3277            |
| 6   | CE             | 1313            |
| 7   | DF             | 2075            |
| 8   | ES             | 1964            |
| 9   | GO             | 1952            |
| 10  | MA             | 726             |

### Insights:

- The top 3 states are SP, RJ, MG which account for more than 60% of total customers
- RR, AP, AC, AM, RO are bottom 5 states with only **~0.57%** of the total customers.

### Action plan:

- Drive multiple marketing strategies in bottom level states.
- Retention programs to be run in top states for maximum customer retention.

#### 4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

- I. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

Query:

```
select a.month,
       a.total_cost_of_orders_in_2017,
       b.total_cost_of_orders_in_2018,
       round((b.total_cost_of_orders_in_2018-
a.total_cost_of_orders_in_2017)/a.total_cost_of_orders_in_2017,2) as
percentage_increament_in_orders
from (
  select format_date('%B', a.order_purchase_timestamp) as month,
         round(sum(b.payment_value), 2) as total_cost_of_orders_in_2017
  from `target.orders` a inner join `target.payments` b on a.order_id =
b.order_id
  where extract(year from a.order_purchase_timestamp) = 2017 and
        extract(month from a.order_purchase_timestamp) between 1 and 8
  group by month) as a

join

  (select format_date('%B', a.order_purchase_timestamp) as month,
         round(sum(b.payment_value), 2) as total_cost_of_orders_in_2018
  from `target.orders` a inner join `target.payments` b on a.order_id =
b.order_id
  where extract(year from a.order_purchase_timestamp) = 2018 and
        extract(month from a.order_purchase_timestamp) between 1 and 8
  group by month) as b
on a.month = b.month
order by parse_date('%B',month)
```

Output Screenshot:

| Row | month    | total_cost_of_orders_in_2017 | total_cost_of_orders_in_2018 | percentage_increament_in_orders |
|-----|----------|------------------------------|------------------------------|---------------------------------|
| 1   | January  | 138488.04                    | 1115004.18                   | 7.05                            |
| 2   | February | 291908.01                    | 992463.34                    | 2.4                             |
| 3   | March    | 449863.6                     | 1159652.12                   | 1.58                            |
| 4   | April    | 417788.03                    | 1160785.48                   | 1.78                            |
| 5   | May      | 592918.82                    | 1153982.15                   | 0.95                            |
| 6   | June     | 511276.38                    | 1023880.5                    | 1.0                             |
| 7   | July     | 592382.92                    | 1066540.75                   | 0.8                             |
| 8   | August   | 674396.32                    | 1022425.32                   | 0.52                            |

### Insights:

- i. Every month from Jan to Aug shows positive growth in total order cost.
- ii. January shows the highest percentage increase of 7.05%
- iii. Growth percentage declines month by month from 7.05% in January to just 0.52% in August

### Action plan:

- i. Since January has the highest growth, identify what drove the spike and replicate similar in other months if applicable.

## II. Calculate the Total & Average value of order price for each state.

### Query:

```
select a.customer_state,
       round(sum(c.payment_value),2) as Total_value_of_order,
       round(avg(c.payment_value),2) as Average_value_of_order
from `target.customers` a left join `target.orders` b on a.customer_id =
b.customer_id inner join `target.payments` c on b.order_id = c.order_id
group by 1
order by 2 desc, 3 desc
```

### Output Screenshot:

| Row | customer_state | Total_value_of_order | Average_value_of_order |
|-----|----------------|----------------------|------------------------|
| 1   | SP             | 5998226.96           | 137.5                  |
| 2   | RJ             | 2144379.69           | 158.53                 |
| 3   | MG             | 1872257.26           | 154.71                 |
| 4   | RS             | 890898.54            | 157.18                 |
| 5   | PR             | 811156.38            | 154.15                 |
| 6   | SC             | 623086.43            | 165.98                 |
| 7   | BA             | 616645.82            | 170.82                 |
| 8   | DF             | 355141.08            | 161.13                 |
| 9   | GO             | 350092.31            | 165.76                 |
| 10  | ES             | 325967.55            | 154.71                 |

### Insights:

- i. State SP leads by huge margin in total order value.
- ii. States like RO, AP, AC, RR, CE, PE and AM exhibit relatively high average order values, suggesting that individual purchases tend to be of higher value.
- iii. States like PB, RO and AP have high average order values but lower total values, represent opportunities for potential growth.

### Action plan:

- i. Run customer acquisition campaigns in lower total order states to increase volume



### III. Calculate the Total & Average value of order freight for each state.

Query:

```
select a.customer_state,  
       round(sum(c.freight_value),2) as Total_freight_value,  
       round(avg(c.freight_value),2) as Average_freight_value  
from `target.customers` a left join `target.orders` b on a.customer_id =  
b.customer_id inner join `target.order_items` c on b.order_id =  
c.order_id  
group by 1  
order by 2 desc, 3 desc
```

Output Screenshot:

| Row | customer_state | Total_freight_value | Average_freight_value |
|-----|----------------|---------------------|-----------------------|
| 1   | SP             | 718723.07           | 15.15                 |
| 2   | RJ             | 305589.31           | 20.96                 |
| 3   | MG             | 270853.46           | 20.63                 |
| 4   | RS             | 135522.74           | 21.74                 |
| 5   | PR             | 117851.68           | 20.53                 |
| 6   | BA             | 100156.68           | 26.36                 |
| 7   | SC             | 89660.26            | 21.47                 |
| 8   | PE             | 59449.66            | 32.92                 |
| 9   | GO             | 53114.98            | 22.77                 |
| 10  | DF             | 50625.5             | 21.04                 |

Insights:

- States like SP, RJ, MG clearly comes with the highest total freight value.
- States with lower average freight value like PI, MA, PB and AL reduces the freight cost.
- States with highest average freight value could imply longer distances or less developed infrastructure.
- States with very high total freight volumes tend to have lower average freight values and vice versa.

Action plan:

- For lower freight cost states, highlight low shipping costs in marketing to attract more customers

## 5. Analysis based on sales, freight and delivery time.

- I. Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Query:

```
select order_id,  
       order_purchase_timestamp,  
       order_delivered_customer_date,  
       order_estimated_delivery_date,  
       date_diff(order_delivered_customer_date, order_purchase_timestamp,  
day) as time_to_deliver,  
       date_diff(order_estimated_delivery_date, order_delivered_customer_  
date, day) as diff_estimated_delivery  
from `target.orders`
```

Output Screenshot:

| Row | order_id                          | order_purchase_timestamp | order_delivered_customer_date | order_estimated_delivery_date | time_to_deliver | diff_estimated_delivery |
|-----|-----------------------------------|--------------------------|-------------------------------|-------------------------------|-----------------|-------------------------|
| 1   | 65d1e226dfaeb8cdc42f665422522d14  | 2016-10-03 21:01:41 UTC  | 2016-11-08 10:58:34 UTC       | 2016-11-25 00:00:00 UTC       | 35              | 16                      |
| 2   | 2c45c33d2f9cb8ff8b1c86cc28c11c30  | 2016-10-09 15:39:56 UTC  | 2016-11-09 14:53:50 UTC       | 2016-12-08 00:00:00 UTC       | 30              | 28                      |
| 3   | 1950d777989f6a877539f53795b4c3c3  | 2018-02-19 19:48:52 UTC  | 2018-03-21 22:03:51 UTC       | 2018-03-09 00:00:00 UTC       | 30              | -12                     |
| 4   | bfbdf0f9bdef84302105ad712db648a6c | 2016-09-15 12:16:38 UTC  | 2016-11-09 07:47:38 UTC       | 2016-10-04 00:00:00 UTC       | 54              | -36                     |
| 5   | 98974b076b01553d49ee6467905675a7  | 2016-10-04 15:44:44 UTC  | 2016-11-17 08:38:57 UTC       | 2016-11-24 00:00:00 UTC       | 43              | 6                       |
| 6   | c4b41c36dd589e901f6879f25a74ec1d  | 2016-10-04 12:53:17 UTC  | 2016-11-09 13:37:38 UTC       | 2016-11-24 00:00:00 UTC       | 36              | 14                      |
| 7   | d2292ff2201e74c5db154d1b7ae68cbb  | 2016-10-04 16:05:29 UTC  | 2016-11-03 10:08:51 UTC       | 2016-11-24 00:00:00 UTC       | 29              | 20                      |
| 8   | 95e01270fcb9e9863423400103359279  | 2016-10-04 18:52:56 UTC  | 2016-11-04 17:54:00 UTC       | 2016-11-24 00:00:00 UTC       | 30              | 19                      |
| 9   | ed8c7b1b3eb256c70ce0c74231e1da88  | 2016-10-04 09:59:03 UTC  | 2016-11-18 08:51:07 UTC       | 2016-11-24 00:00:00 UTC       | 44              | 5                       |
| 10  | 5cc475c7c03290048eb2e742cd64cb5e  | 2016-10-04 21:54:52 UTC  | 2016-12-12 20:31:54 UTC       | 2016-11-24 00:00:00 UTC       | 68              | -18                     |

Insights:

- i. Average delivery time is ~37 days.
- ii. Most orders fall between 29 and 41 days for delivery
- iii. 75% of order delivered were delivered late than estimated.

Action plan:

- i. Identify areas with fast delivery performance and apply similar strategies in low delivery areas

## II. Find out the top 5 states with the highest & lowest average freight value.

Query:

-- Query for top 5 states with highest avg freight value

```
select a.customer_state,  
       round(avg(c.freight_value),2) as Average_freight_value  
from `target.customers` a left join `target.orders` b on a.customer_id =  
b.customer_id inner join `target.order_items` c on b.order_id =  
c.order_id  
group by 1  
order by 2 desc  
limit 5
```

-- Query for bottom 5 states with lowest avg freight value

```
select a.customer_state,  
       round(avg(c.freight_value),2) as Average_freight_value  
from `target.customers` a left join `target.orders` b on a.customer_id =  
b.customer_id inner join `target.order_items` c on b.order_id =  
c.order_id  
group by 1  
order by 2 asc  
limit 5
```

Output Screenshot:

| Row                                         | customer_state | Average_freight_value | Row                                           | customer_state | Average_freight_value |
|---------------------------------------------|----------------|-----------------------|-----------------------------------------------|----------------|-----------------------|
| 1                                           | RR             | 42.98                 | 1                                             | SP             | 15.15                 |
| 2                                           | PB             | 42.72                 | 2                                             | PR             | 20.53                 |
| 3                                           | RO             | 41.07                 | 3                                             | MG             | 20.63                 |
| 4                                           | AC             | 40.07                 | 4                                             | RJ             | 20.96                 |
| 5                                           | PI             | 39.15                 | 5                                             | DF             | 21.04                 |
| Top 5 states with highest avg freight value |                |                       | Bottom 5 states with lowest avg freight value |                |                       |

Insights:

- States like RR,PB,RO,AC, PI comes with highest average freight value.
- This could be due to longer distances or less developed infrastructure.
- States like SP,PR,MG,RJ,DF provides advantage of lowest average freight value.

Action plan:

- For lower freight cost states, highlight low shipping costs in marketing to attract more customers

### III. Find out the top 5 states with the highest & lowest average delivery time.

Query:

-- Query for top 5 states with highest average delivery time

```
select a.customer_state,  
       round(avg(date_diff(order_delivered_customer_date,order_purchase_  
timestamp,day)),2) as avg_time_to_deliver  
from `target.customers` a left join `target.orders` b on a.customer_id =  
b.customer_id  
group by 1  
order by 2 desc  
limit 5
```

-- Query for bottom 5 states with lowest average delivery time

```
select a.customer_state,  
       round(avg(date_diff(order_delivered_customer_date,order_purchase_  
timestamp,day)),2) as avg_time_to_deliver  
from `target.customers` a left join `target.orders` b on a.customer_id =  
b.customer_id  
group by 1  
order by 2 asc  
limit 5
```

Output Screenshot:

| Row | customer_state | avg_time_to_deliver | Row | customer_state | avg_time_to_deliver |
|-----|----------------|---------------------|-----|----------------|---------------------|
| 1   | RR             | 28.98               | 1   | SP             | 8.3                 |
| 2   | AP             | 26.73               | 2   | PR             | 11.53               |
| 3   | AM             | 25.99               | 3   | MG             | 11.54               |
| 4   | AL             | 24.04               | 4   | DF             | 12.51               |
| 5   | PA             | 23.32               | 5   | SC             | 14.48               |

Top 5 states with highest avg delivery time

Bottom 5 states with lowest avg delivery time

Insights:

- States like RR,AP,AM,AL,PA comes with highest average delivery time.
- States like SP,PR,MG,DF,SC provides advantage of lowest average delivery time.

Action plan:

- Identify areas with fast delivery performance and apply similar strategies in low delivery areas

**IV. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.**

Query:

```
select a.customer_state,  
       round(avg(date_diff(order_estimated_delivery_date,order_delivered  
_customer_date,day)),2) as avg_diff_from_estimated_delivery  
from `target.customers` a left join `target.orders` b on a.customer_id =  
b.customer_id  
where  
date_diff(order_estimated_delivery_date,order_delivered_customer_date,da  
y) is not null  
group by 1  
order by 2 desc  
limit 5
```

Output Screenshot:

| Row | customer_state | avg_diff_from_estimated_delivery |
|-----|----------------|----------------------------------|
| 1   | AC             | 19.76                            |
| 2   | RO             | 19.13                            |
| 3   | AP             | 18.73                            |
| 4   | AM             | 18.61                            |
| 5   | RR             | 16.41                            |

Insights:

- AC,RO,AP,AM,RR are states which enables very fast delivery for orders.
- This suggests that these states having efficient ways and modes of delivery.
- AC has the highest average difference between estimated date and delivery date at 19.76 days.

Action plan:

- Promote efficient delivery achievements effectively in marketing.

## 6. Analysis based on the payments:

### I. Find the month-on-month no. of orders placed using different payment types.

Query:

```
select format_date('%B - %Y',a.order_purchase_timestamp) as Order_month,
       sum(case when b.payment_type = 'credit_card' then 1 else 0 end)
as Credit_card_payment,
       sum(case when b.payment_type = 'UPI' then 1 else 0 end) as
UPI_payment,
       sum(case when b.payment_type = 'voucher' then 1 else 0 end) as
voucher_payment,
       sum(case when b.payment_type = 'debit_card' then 1 else 0 end) as
debit_card_payment,
       sum(case when b.payment_type = 'not_defined' then 1 else 0 end)
as not_defined_method
from `target.orders` a left join `target.payments` b on
a.order_id=b.order_id
group by 1,
        extract(month from order_purchase_timestamp),
        extract(year from order_purchase_timestamp)
order by extract(year from order_purchase_timestamp),
        extract(month from order_purchase_timestamp)
```

Output Screenshot:

| Row | Order_month      | Credit_card_payment | UPI_payment | voucher_payment | debit_card_payment | not_defined_method |
|-----|------------------|---------------------|-------------|-----------------|--------------------|--------------------|
| 1   | September - 2016 | 3                   | 0           | 0               | 0                  | 0                  |
| 2   | October - 2016   | 254                 | 63          | 23              | 2                  | 0                  |
| 3   | December - 2016  | 1                   | 0           | 0               | 0                  | 0                  |
| 4   | January - 2017   | 583                 | 197         | 61              | 9                  | 0                  |
| 5   | February - 2017  | 1356                | 398         | 119             | 13                 | 0                  |
| 6   | March - 2017     | 2016                | 590         | 200             | 31                 | 0                  |
| 7   | April - 2017     | 1846                | 496         | 202             | 27                 | 0                  |
| 8   | May - 2017       | 2853                | 772         | 289             | 30                 | 0                  |
| 9   | June - 2017      | 2463                | 707         | 239             | 27                 | 0                  |
| 10  | July - 2017      | 3086                | 845         | 364             | 22                 | 0                  |

Insights:

- Significant increase in overall transaction volumes from 2016 to 2018.
- With Credit card and UPI being the most popular methods, followed by the significant rise of voucher payments.
- Maximum of orders were processed through credit card payment which comes as favorite mode of transaction.

Action plan:

- Focus on strengthening primary payment modes like credit card and UPI.
- Promote safety in all payment modes.

**II. Find the no. of orders placed on the basis of the payment installments that have been paid.**

Query:

```
select payment_installments,  
       count(*) as total_orders  
from `target.payments`  
group by 1  
order by 1
```

Output Screenshot:

| Row | payment_installments | total_orders |
|-----|----------------------|--------------|
| 1   | 0                    | 2            |
| 2   | 1                    | 52546        |
| 3   | 2                    | 12413        |
| 4   | 3                    | 10461        |
| 5   | 4                    | 7098         |
| 6   | 5                    | 5239         |
| 7   | 6                    | 3920         |
| 8   | 7                    | 1626         |
| 9   | 8                    | 4268         |
| 10  | 9                    | 644          |

Insights:

- Decreasing trend in total orders as the number of payment installments increase.
- Highest number of orders were placed with 1 installment.
- This suggests a peak preference for single-installment or very short-term installment options.

Action plan:

- Offer multiple payment option and offers for more number of installments.